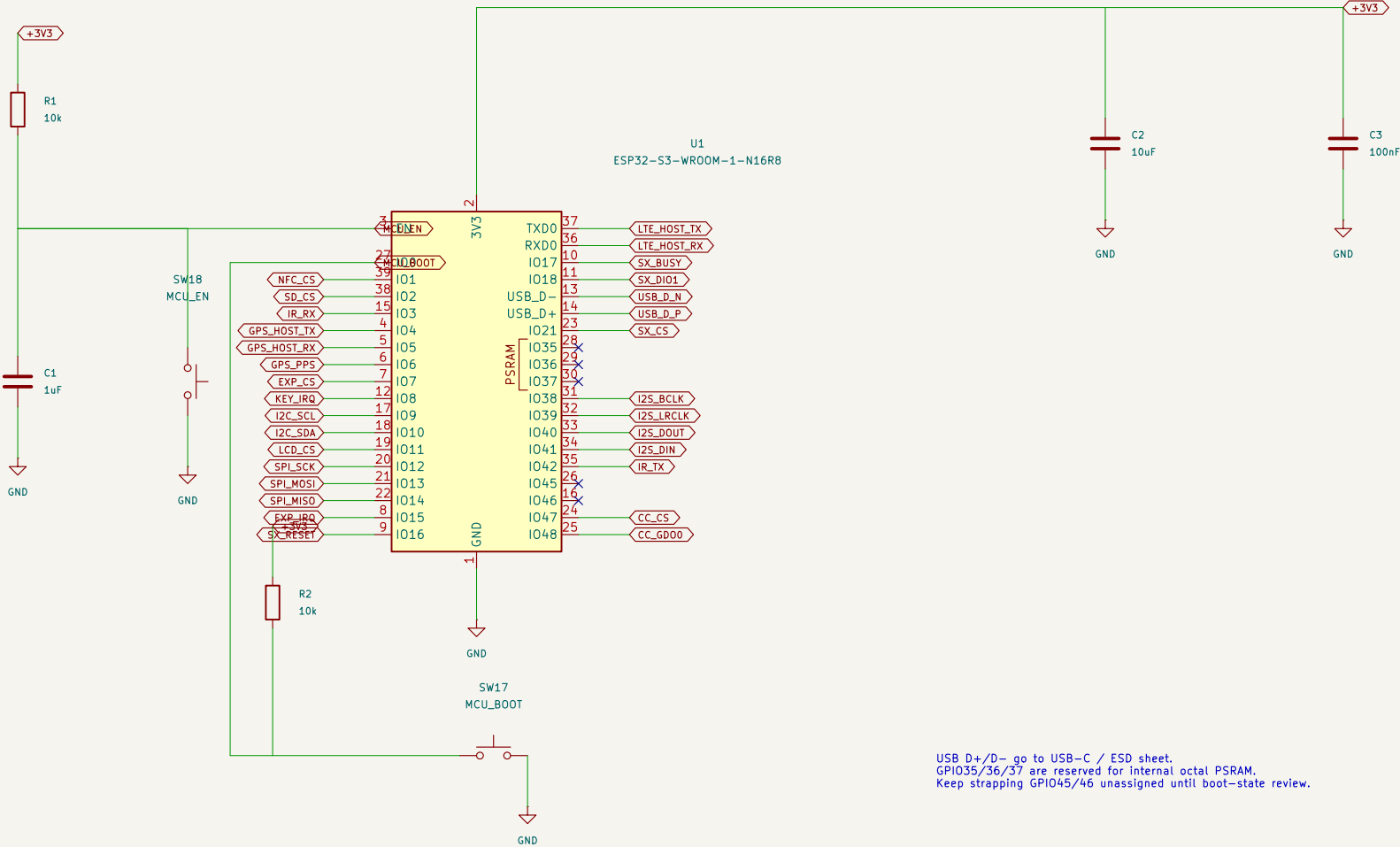




ESP32 host, boot and reset

Retain N16R8. Module pads 28/29/30 belong to octal PSRAM. USB programming uses the USB sheet.



USB D+/D- go to USB-C / ESD sheet.  
GPIO35/36/37 are reserved for internal octal PSRAM.  
Keep strapping GPIO45/46 unassigned until boot-state review.

RESET / EN: 10k pull-up + 1uF  
BOOT held low during reset selects download mode.

Circuit review in progress – not released for fabrication

Sheet: /ESP32 host, boot and reset/  
File: core.kicad\_sch

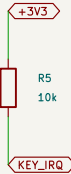
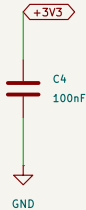
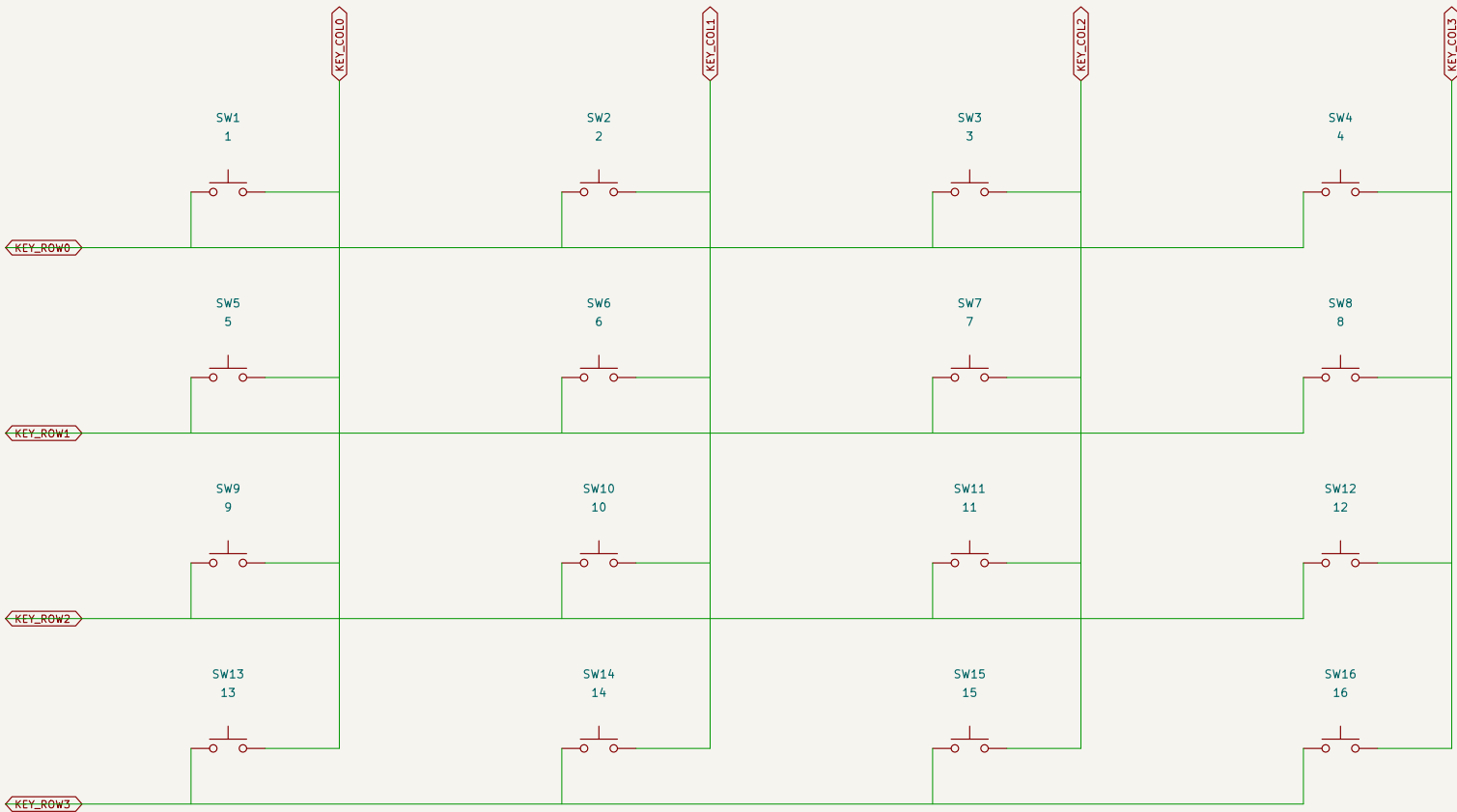
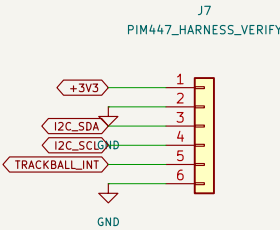
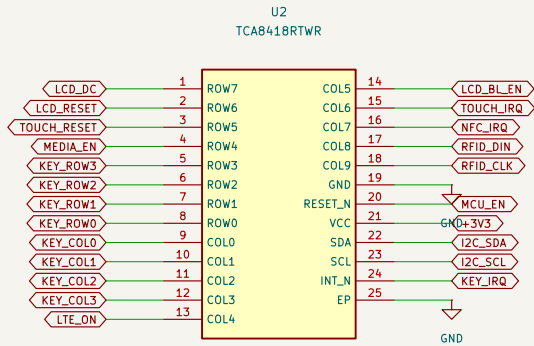
**Title: ESP32 host, boot and reset**

Size: A3  
Date: 2026-09-24

Rev: A – engineering draft  
Id: 2/23

Key matrix and trackball

TCAB418: 4 rows x 4 columns; remaining pins are firmware-configured GPIO. Confirm PIM447 harness orientation.



Circuit review in progress – not released for fabrication

Sheet: /Key matrix and trackball/  
File: keys.kicad\_sch

**Title: Key matrix and trackball**

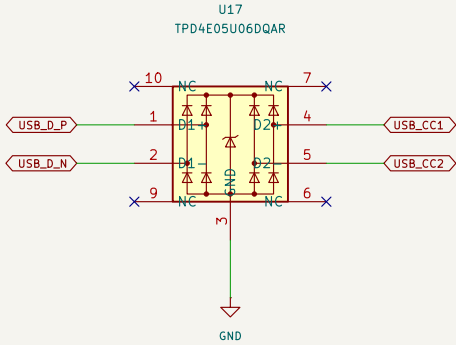
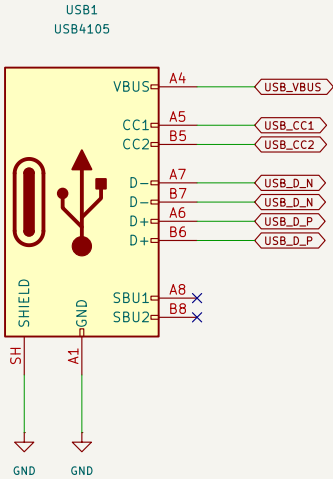
Size: A3  
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft  
Id: 3/23

USB-C data and ESD

CC1/CC2 terminate at the sink controller on usb-detect. Data pair goes to the ESP32. Shield/ground return needs layout review.



Circuit review in progress – not released for fabrication

Sheet: /USB-C data and ESD/  
File: usb.kicad\_sch

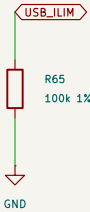
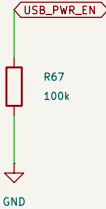
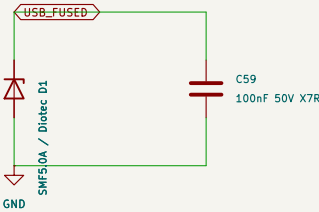
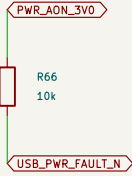
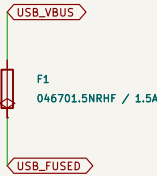
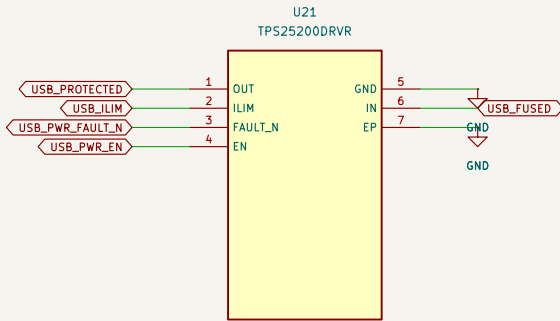
**Title: USB-C data and ESD**

Size: A3 Date: 2026-09-24  
KiCad E.D.A. 10.0.6

Rev: A – engineering draft  
Id: 4/23

Fused USB input and eFuse

TPS25200 R\_ILIM = 100k, 1%. F1/TVS pulse coordination requires bench validation. No USB PD negotiation.



Supply flag: external USB VBUS through F1. No internal source is implied.

Circuit review in progress – not released for fabrication

Sheet: /Fused USB input and eFuse/  
File: usb-input.kicad\_sch

Title: **Fused USB input and eFuse**

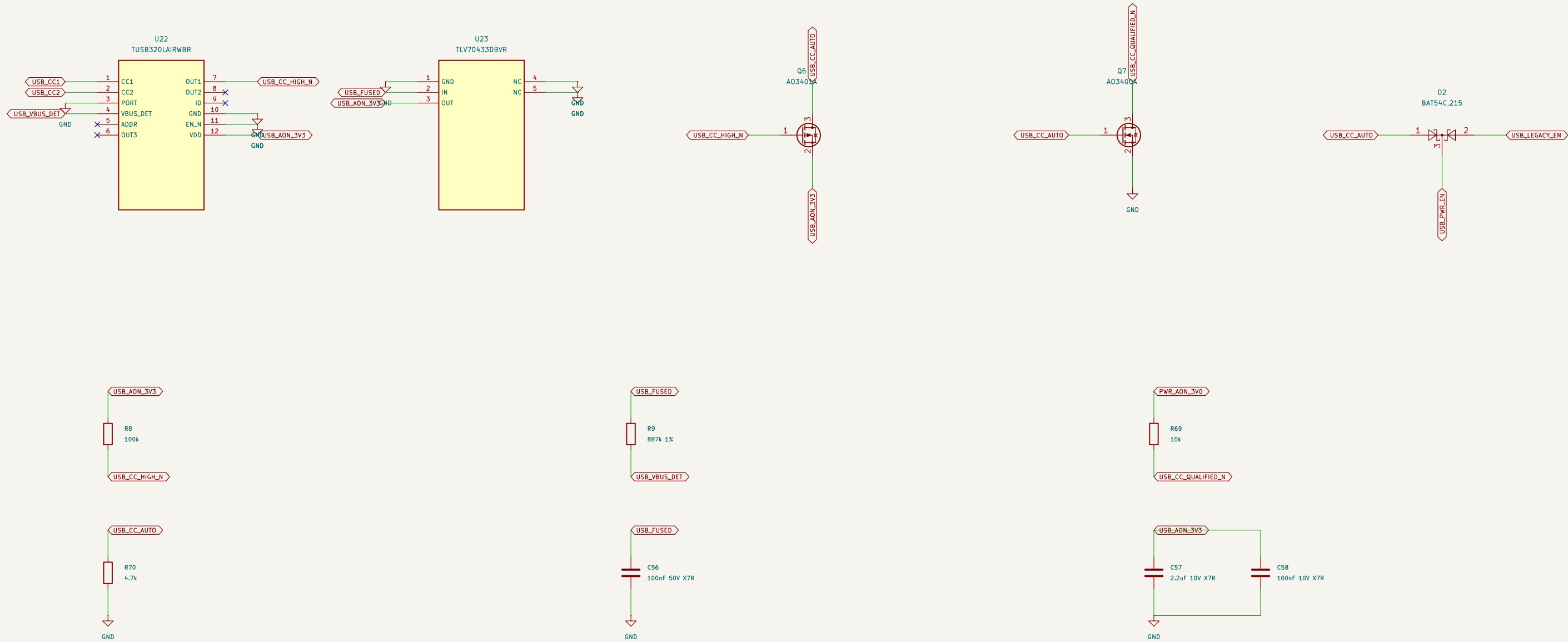
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KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: **A – engineering draft**  
Id: 5/23

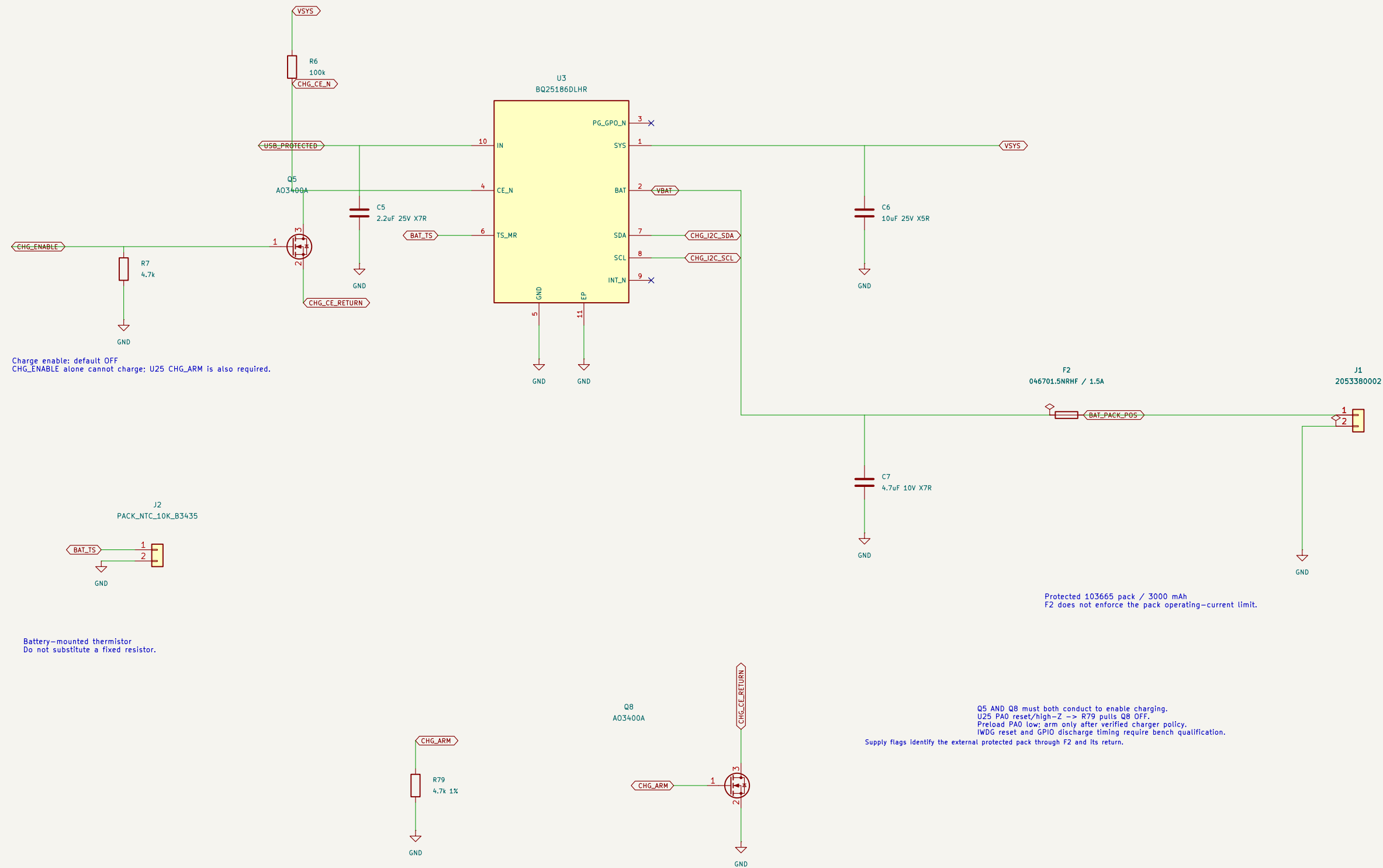
USB-C source detection and permission

Sink-only internal Rd. OUT1 low authorizes Type-C 1.5A/3A. Legacy permission requires USB enumeration; see charger-policy.json.



Battery, thermistor and programmable charger

Verify actual battery polarity. J2 is a battery-mounted 10k B3435 NTC. CE defaults OFF. Firmware must apply charger-policy.json.



Protected 103665 pack / 3000 mAh  
F2 does not enforce the pack operating-current limit.

Q5 AND Q8 must both conduct to enable charging.  
U25 PA0 reset/high-Z --> R79 pulls Q8 OFF.  
Preload PA0 low; arm only after verified charger policy.  
IWDG reset and GPIO discharge timing require bench qualification.

Supply flags identify the external protected pack through F2 and its return.

Circuit review in progress – not released for fabrication

Sheet: /Battery, thermistor and programmable charger/  
File: power.kicad\_sch

**Title: Battery, thermistor and programmable charger**

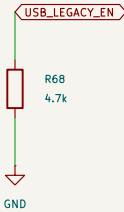
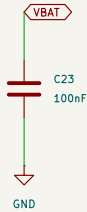
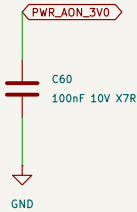
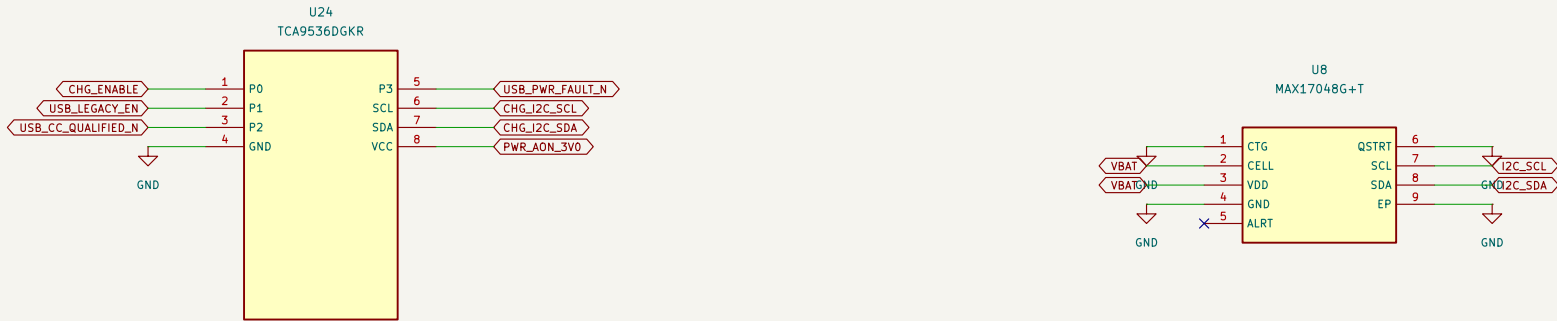
|          |                  |
|----------|------------------|
| Size: A3 | Date: 2026-09-24 |
|----------|------------------|

Rev: A – engineering draft

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Power control and fuel gauge

I2C: TCA9536 0x41, MAX17048 0x36, charger 0x6A. Preload low output latches before enabling charger—control outputs.



Circuit review in progress – not released for fabrication

Sheet: /Power control and fuel gauge/  
File: power-control.kicad\_sch

**Title: Power control and fuel gauge**

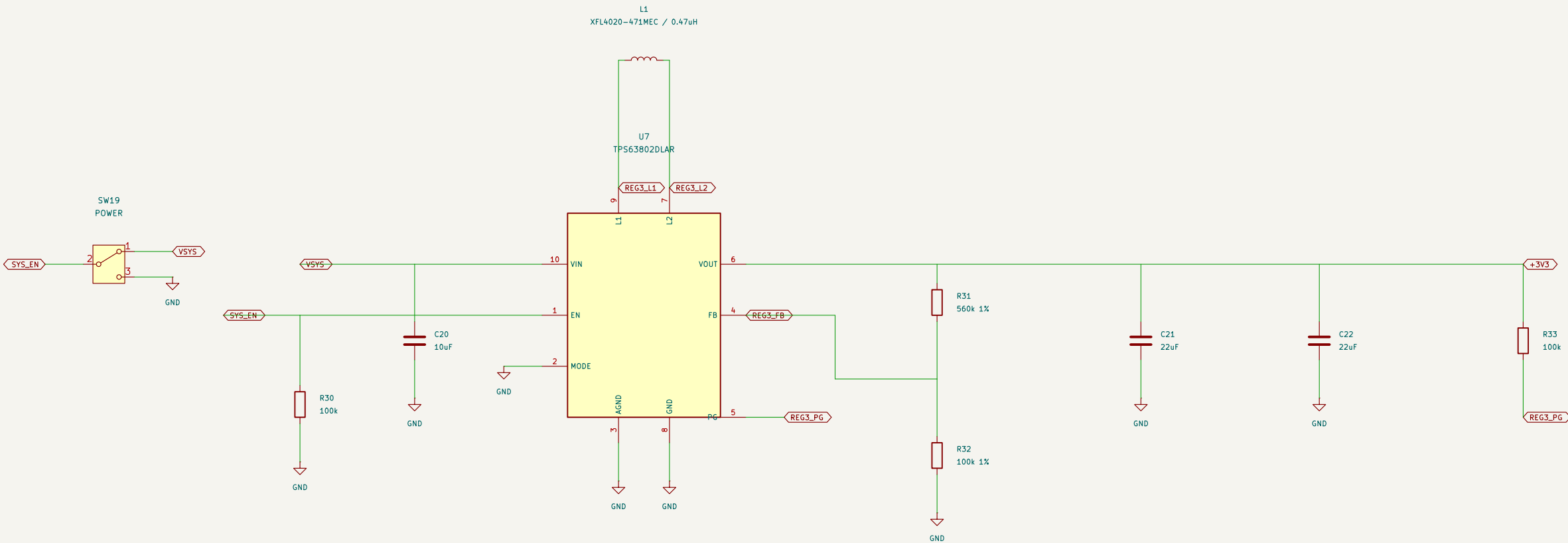
Size: A3  
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft  
Id: 8/23

Main 3.3 V buck–boost supply

TPS63802 local PCB copper is routed. External distribution, effective capacitance, thermal limits and load transients remain to qualify.



0.6 V reference x (1 + 560k / 100k) = 3.3 V nominal.  
Feedback senses the output capacitor node; keep it clear of the switching loop.

SW19 controls SYS\_EN.  
Battery / charger circuitry can remain energized while logic rails are off.

Circuit review in progress – not released for fabrication

Sheet: /Main 3.3 V buck–boost supply/  
File: reg3.kicad\_sch

**Title: Main 3.3 V buck–boost supply**

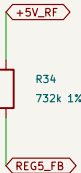
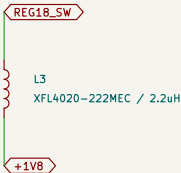
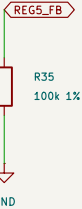
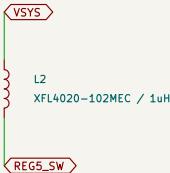
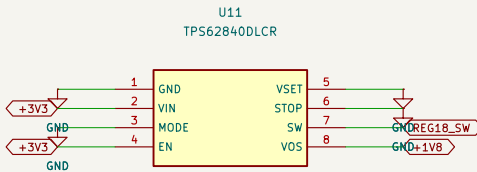
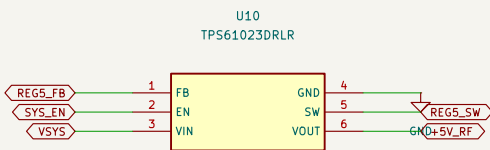
Size: A3  
KiCad E.D.A. 10.0.6

Date: 2026–09–24

Rev: A – engineering draft  
Id: 9/23

Auxiliary 5 V and 1.8 V supplies

Switching loops must be short in PCB layout. The 5 V rail also supplies display backlight and the LF reader.



Circuit review in progress – not released for fabrication

Sheet: /Auxiliary 5 V and 1.8 V supplies/  
File: aux-power.kicad\_sch

**Title: Auxiliary 5 V and 1.8 V supplies**

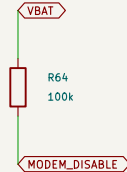
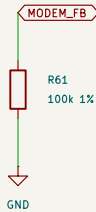
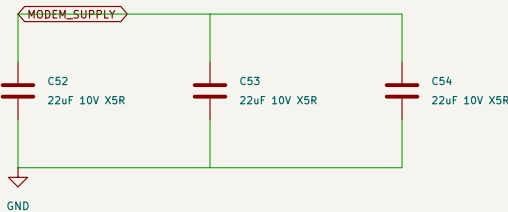
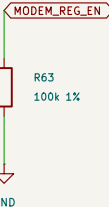
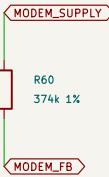
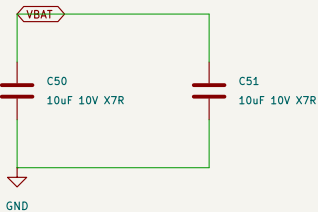
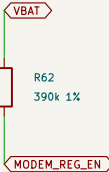
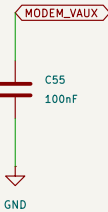
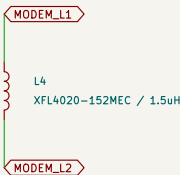
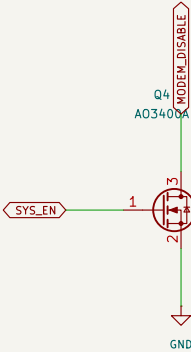
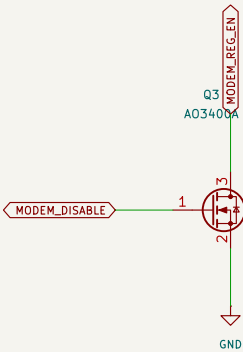
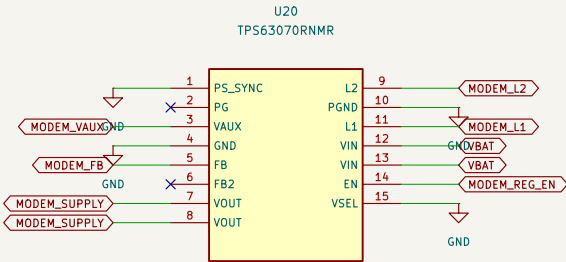
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KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft  
Id: 10/23

Regulated modem supply and enable

Nominal output 3.792 V. Exact modem burst current and MakerFocus pack current budget are unresolved release blockers.



Circuit review in progress – not released for fabrication

Sheet: /Regulated modem supply and enable/  
File: modem-power.kicad\_sch

**Title: Regulated modem supply and enable**

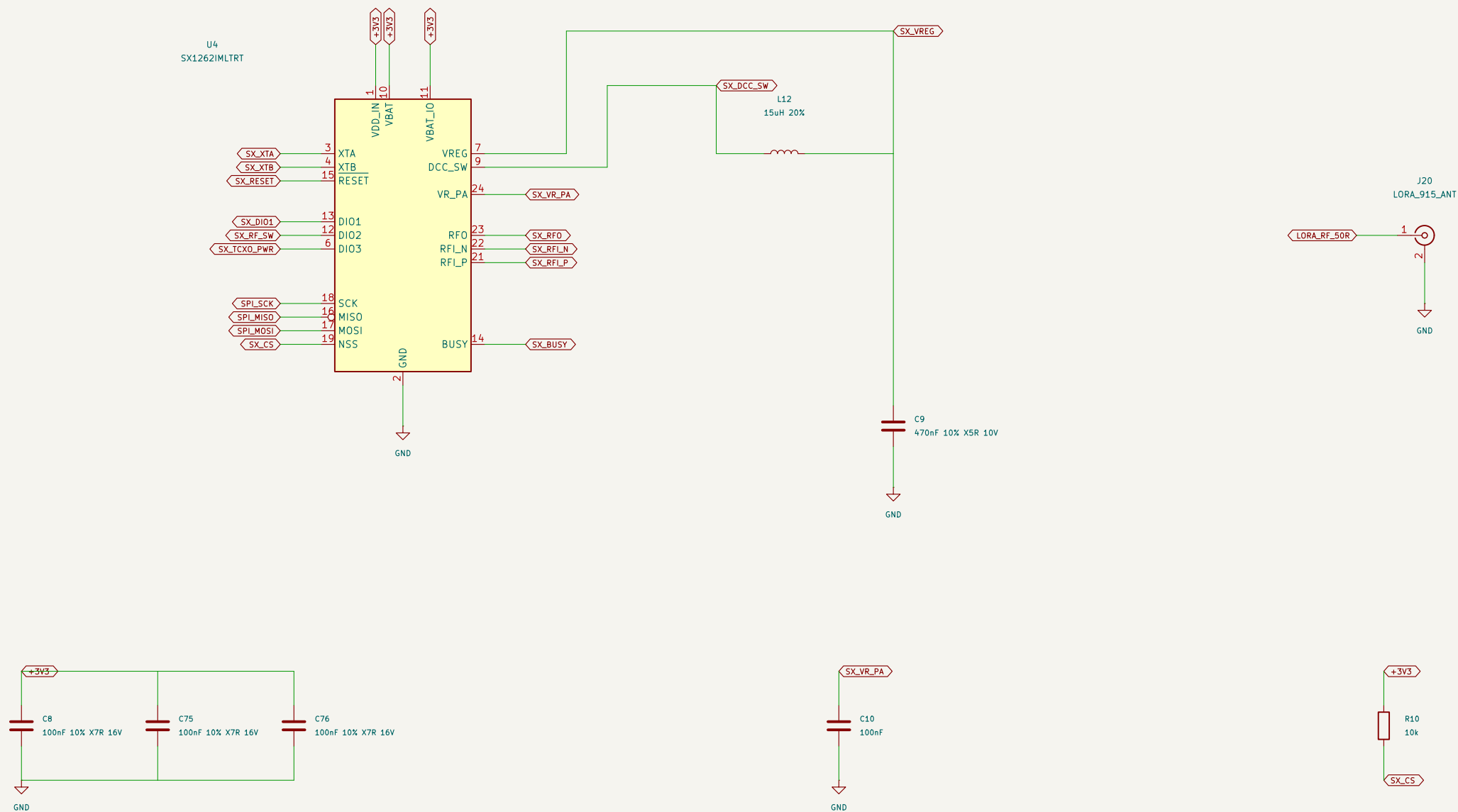
Size: A3  
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft  
Id: 11/23

## SX1262 LoRa radio

DC-DC support captured from Semtech DS rev1.2 Fig14-2 and Table5-3. Clock and RF frontend remain incomplete. Do not power.



C8: VBAT pin10. C75: VDD\_IN pin1. C76: VBAT\_IO pin11.  
All use the same +3V3 rail; place each close to its pin.  
C9 replaces the incorrect 100nF with Semtech reference 470nF.

SetRegulatorMode(0x01) only in STDBY\_RC, after hardware qualification.  
L12 supplies the core, not the SX1262 high-power PA.  
TCXO, PA feed, matching/filter and RF switch are still missing.  
C10 PA bypass and J20 antenna connection remain provisional.

Circuit review in progress – not released for fabrication

Sheet: /SX1262 LoRa radio/  
File: radios.kicad\_sch

**Title: SX1262 LoRa radio**

Size: A3 Date: 2026-09-24

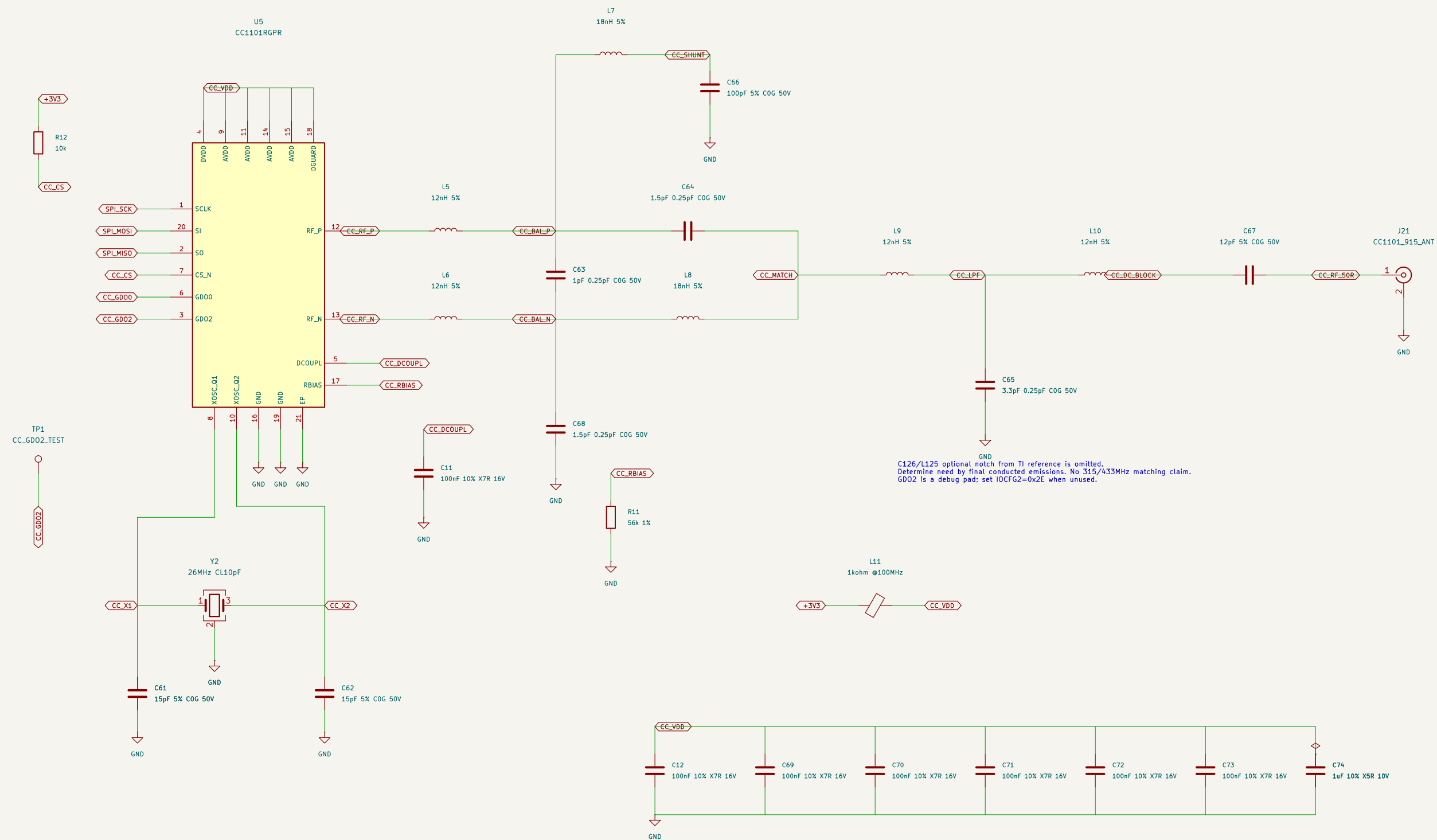
KiCad E.D.A. 10.0.6

Rev: A – engineering draft

Id: 12/23

CC1101 sub-GHz radio

868/915MHz frontend: TI SWRR045 rev3.0 / SWRS061I Fig11. RF layout and tuning are NOT qualified. Separate 50-ohm antenna.



Y2: ABM8-26.000MHZ-10-D-1-G-T  
CL = 15pF / 2 + estimated 2.5pF stray = 10pF.  
Measure oscillator frequency, startup and drive level.

CC\_VDD flag: main +3V3 through passive L11; not an independent source.  
Place one 100nF at each supply pin group, plus local 1uF bulk.

Circuit review in progress – not released for fabrication

Sheet: /CC1101 sub-GHz radio/  
File: cc-radio.kicad\_sch

Title: CC1101 sub-GHz radio

Size: A3 Date: 2026-09-24

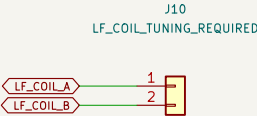
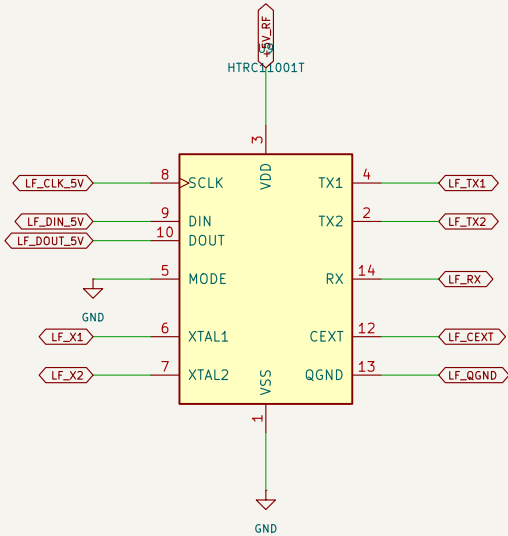
KiCad E.D.A. 10.0.6

Rev: A – engineering draft

Id: 13/23

125 kHz reader analog circuit

HTRC110 requires its oscillator, QGND/CEXT bypass, resonant coil network and a timing-capable host interface.



Circuit review in progress – not released for fabrication

Sheet: /125 kHz reader analog circuit/  
File: lf-radio.kicad\_sch

**Title: 125 kHz reader analog circuit**

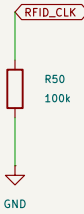
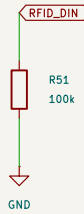
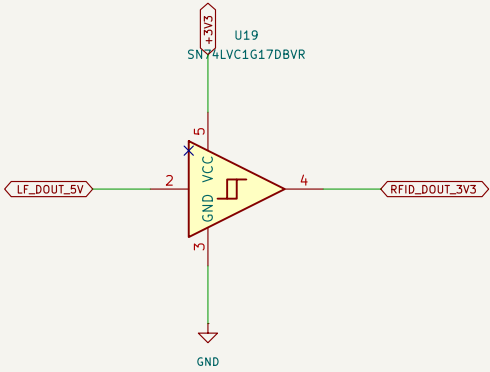
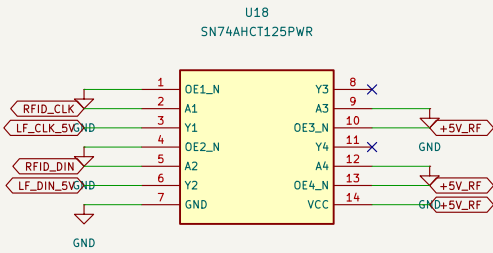
Size: A3  
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft  
Id: 14/23

LF reader logic translation

AHCT inputs accept 3.3 V logic at 5 V supply; LVC return buffer is powered at 3.3 V. Host timing remains a design requirement.



Circuit review in progress – not released for fabrication

Sheet: /LF reader logic translation/  
File: lf-support.kicad\_sch

**Title: LF reader logic translation**

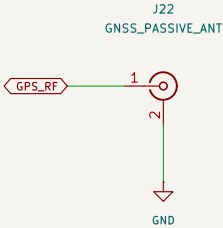
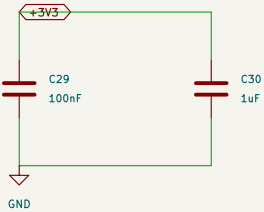
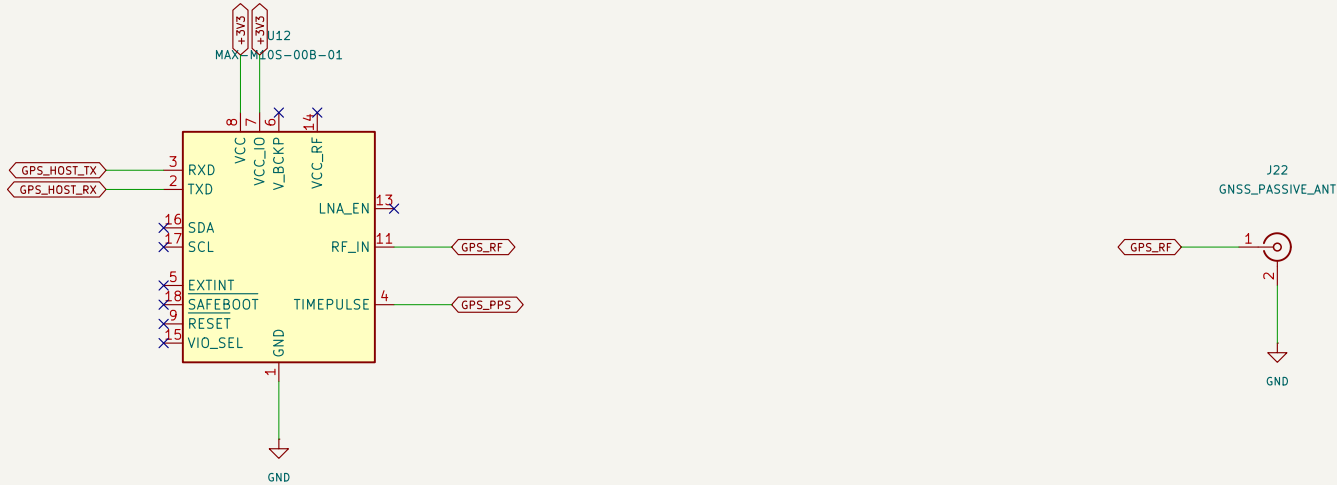
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KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft  
Id: 15/23

GNSS receiver and passive antenna

MAX-M10S passive antenna. V\_BCKP remains open: cold starts after power removal. No direct LiPo connection.



Circuit review in progress – not released for fabrication

Sheet: /GNSS receiver and passive antenna/  
File: gnss-antennas.kicad\_sch

**Title: GNSS receiver and passive antenna**

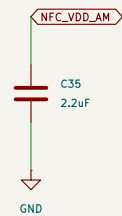
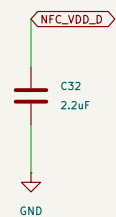
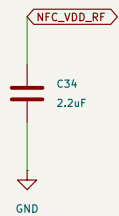
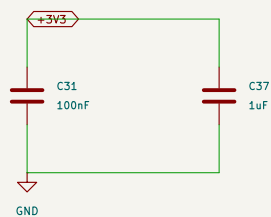
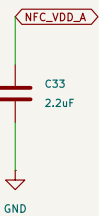
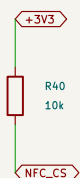
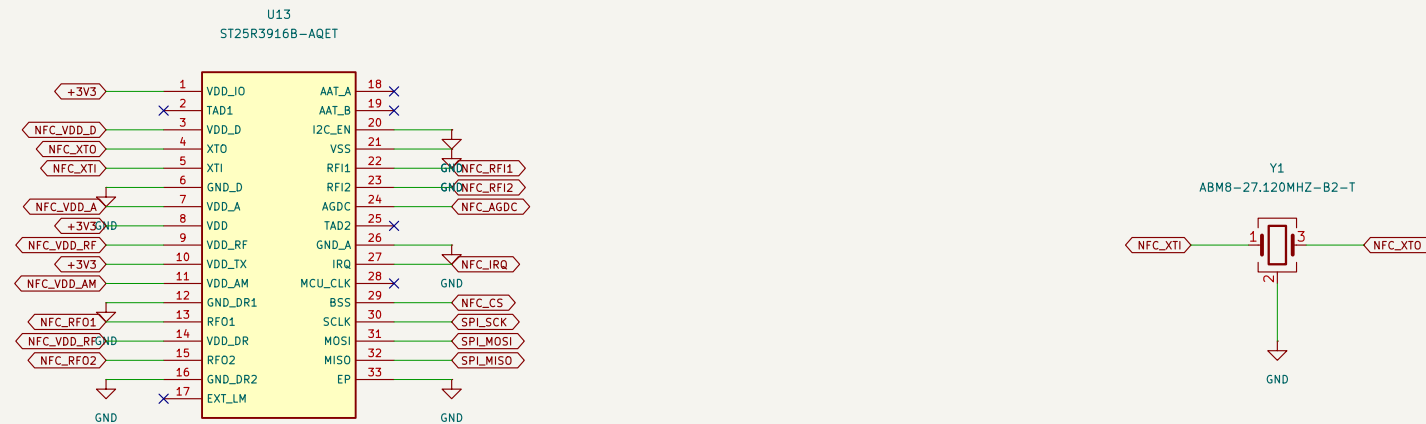
Size: A3  
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft  
Id: 16/23

NFC reader, oscillator and local supplies

ST25R3916B needs a tuned case-mounted loop and TX/RX matching network. Verify crystal load against the selected crystal.



Circuit review in progress – not released for fabrication

Sheet: /NFC reader, oscillator and local supplies/  
File: nfc.kicad\_sch

**Title: NFC reader, oscillator and local supplies**

Size: A3

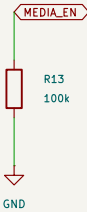
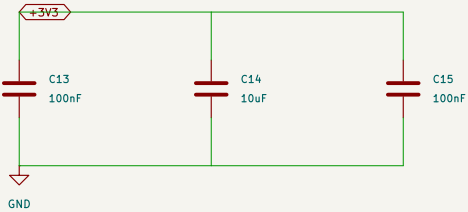
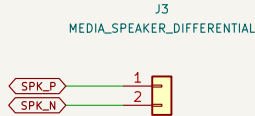
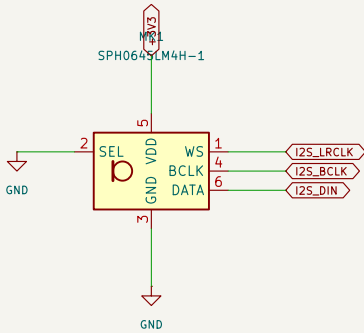
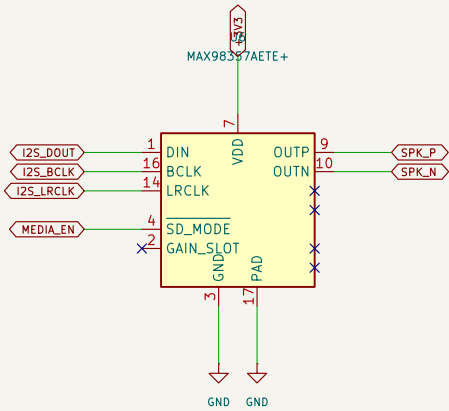
Date: 2026-09-24

Rev: A – engineering draft

Id: 17/23

Media speaker amplifier and digital microphone

Top-mounted media speaker uses differential outputs: neither speaker wire is ground. Digital microphone is case-bottom mounted.



Circuit review in progress – not released for fabrication

Sheet: /Media speaker amplifier and digital microphone/  
File: audio.kicad\_sch

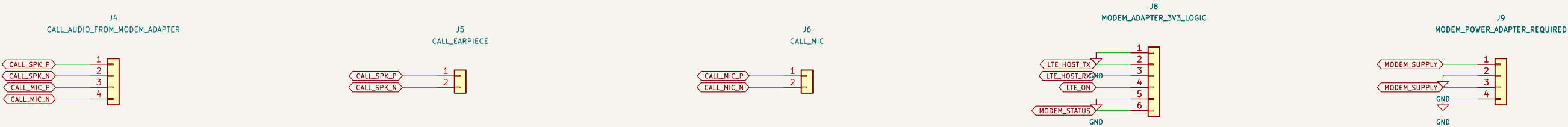
**Title: Media speaker amplifier and digital microphone**

Size: A3  
Date: 2026-09-24

Rev: A – engineering draft  
Id: 18/23

Cellular adapter and call–audio harnesses

Replaceable modem adapter owns SIM, cellular RF and modem–level translation. No universal–carrier guarantee. Speaker top / mic bottom.



Circuit review in progress – not released for fabrication

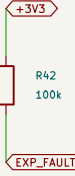
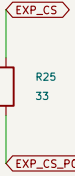
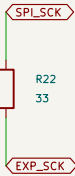
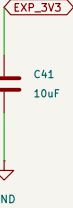
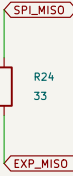
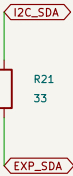
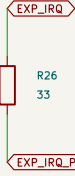
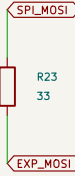
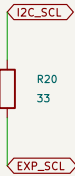
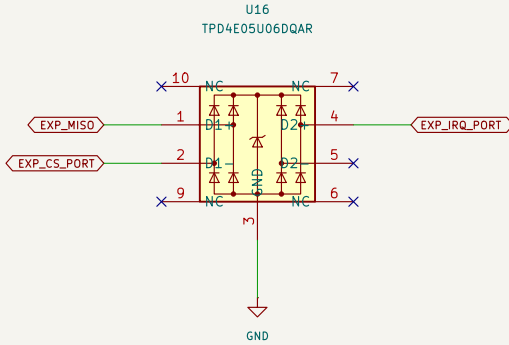
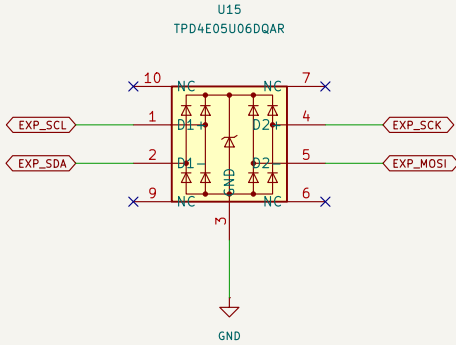
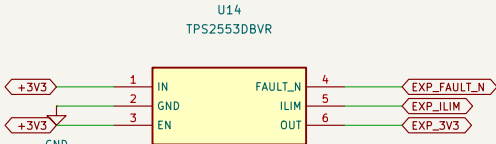
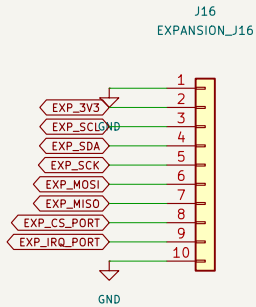
Sheet: /Cellular adapter and call–audio harnesses/  
File: interfaces.kicad\_sch

**Title: Cellular adapter and call–audio harnesses**

|                     |                  |                            |
|---------------------|------------------|----------------------------|
| Size: A3            | Date: 2026–09–24 | Rev: A – engineering draft |
| KiCad E.D.A. 10.0.6 |                  | Id: 19/23                  |

Protected external module port

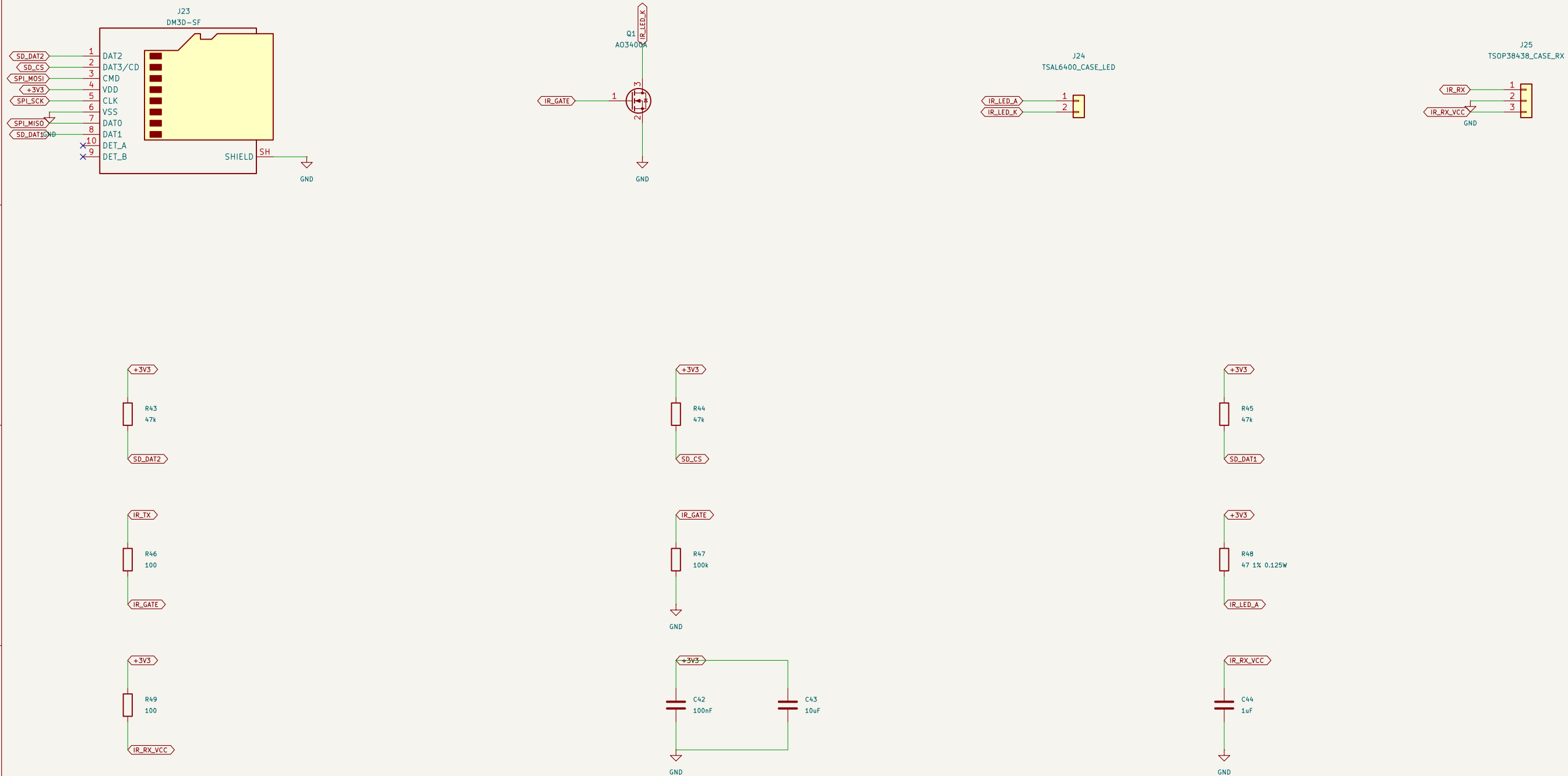
J16 pin order retained. Switch accessories with power off. TPS2553 limit and ESD path require qualification.



|                                                                       |                  |                            |
|-----------------------------------------------------------------------|------------------|----------------------------|
| Circuit review in progress – not released for fabrication             |                  |                            |
| Sheet: /Protected external module port/<br>File: protection.kicad_sch |                  |                            |
| Title: Protected external module port                                 |                  |                            |
| Size: A3                                                              | Date: 2026-09-24 | Rev: A – engineering draft |
| KiCad E.D.A. 10.0.6                                                   |                  | Id: 20/23                  |

MicroSD and case-mounted infrared

SPI microSD, transistor-driven IR LED, filtered receiver supply. Confirm case LED/receiver polarity and GPIO3 boot behavior.



Circuit review in progress – not released for fabrication

Sheet: /MicroSD and case-mounted infrared/  
File: storage-ir.kicad\_sch

Title: MicroSD and case-mounted infrared

Size: A3

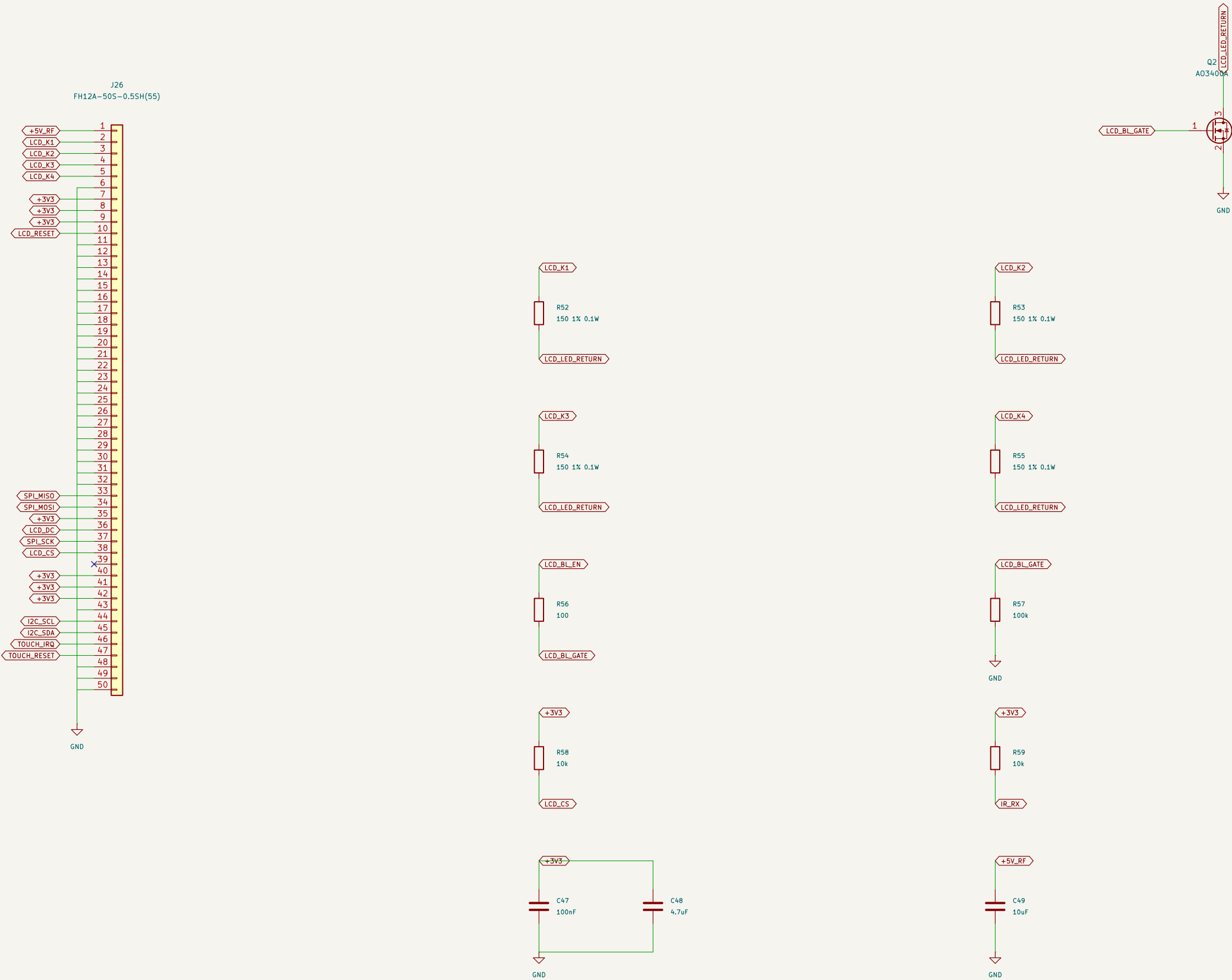
Date: 2026-09-24

Rev: A – engineering draft

Id: 21/23

External display, touch and backlight

ER-TFT024IPS-3 connector contract. Verify purchased CTP option, FPC orientation, LED branches and supply tolerances.



Circuit review in progress – not released for fabrication

Sheet: /External display, touch and backlight/  
File: display.kicad\_sch

**Title: External display, touch and backlight**

Size: A3  
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft  
Id: 22/23

Always-on charging supervisor

Separate supervisor selected. U3 remains the charger. Main host powers off at SW19. Circuit capture only; routing, STM32 firmware and fault qualification pending.

A

B

C

D

E

F

A

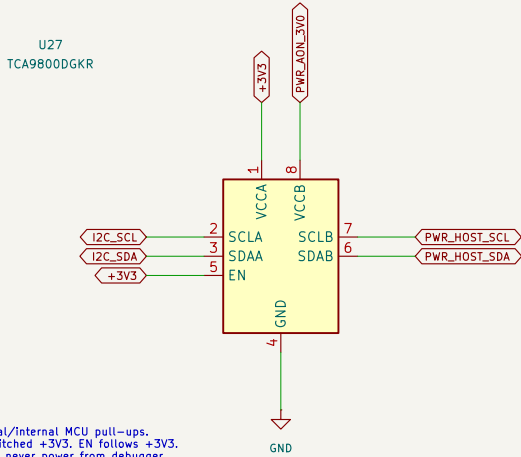
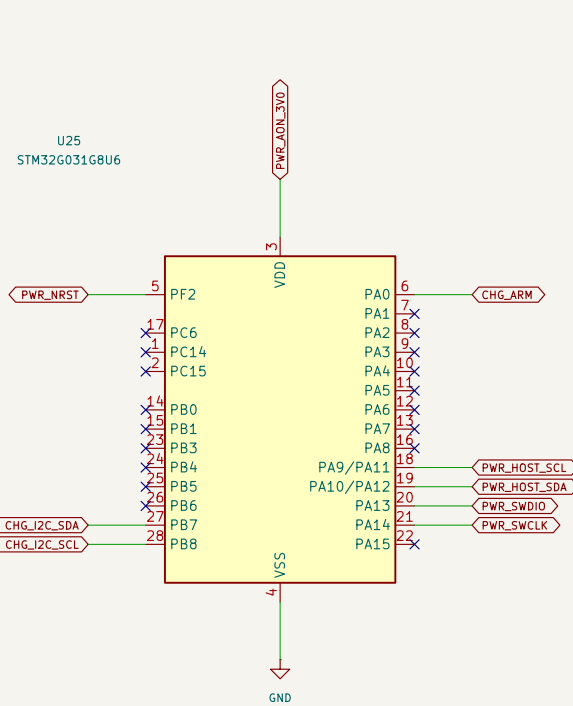
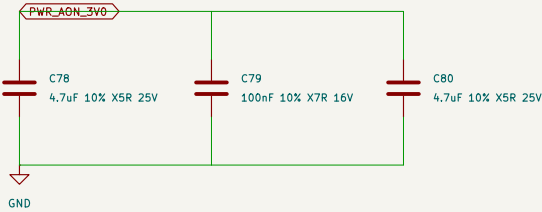
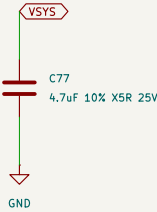
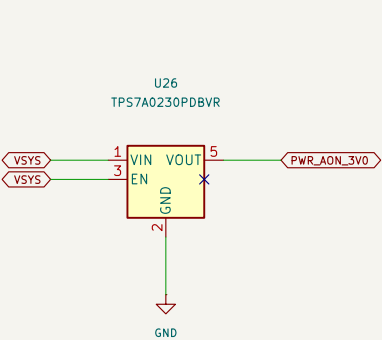
B

C

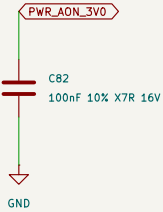
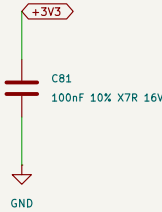
D

E

F

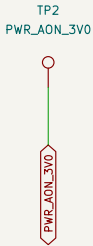
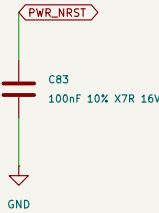
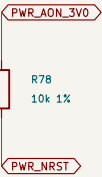
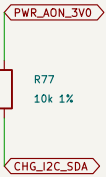
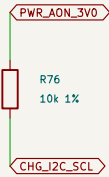


TCA9800 B side: NO external/Internal MCU pull-ups.  
A-side pull-ups are on switched +3V3. EN follows +3V3.  
Debug VTREF is sense only; never power from debugger.



I2C1: PB8/PB7 AF6 -> charger 0x6A + expander 0x41.  
I2C2: remap PA9/PA10 to PA11/PA12, AF6.  
Host slave address 0x42 reserved; protocol not implemented.

PA0: dedicated active-high charge arm. Reset analog/high-Z allows R79 at Q8 to hold charging disabled independently of U24. Never retain this output through standby or run bootloader while armed.



Circuit review in progress – not released for fabrication

Sheet: /Always-on charging supervisor/  
File: power-supervisor.kicad\_sch

Title: Always-on charging supervisor

Size: A3  
Date: 2026-09-24

Rev: A – engineering draft  
Id: 23/23